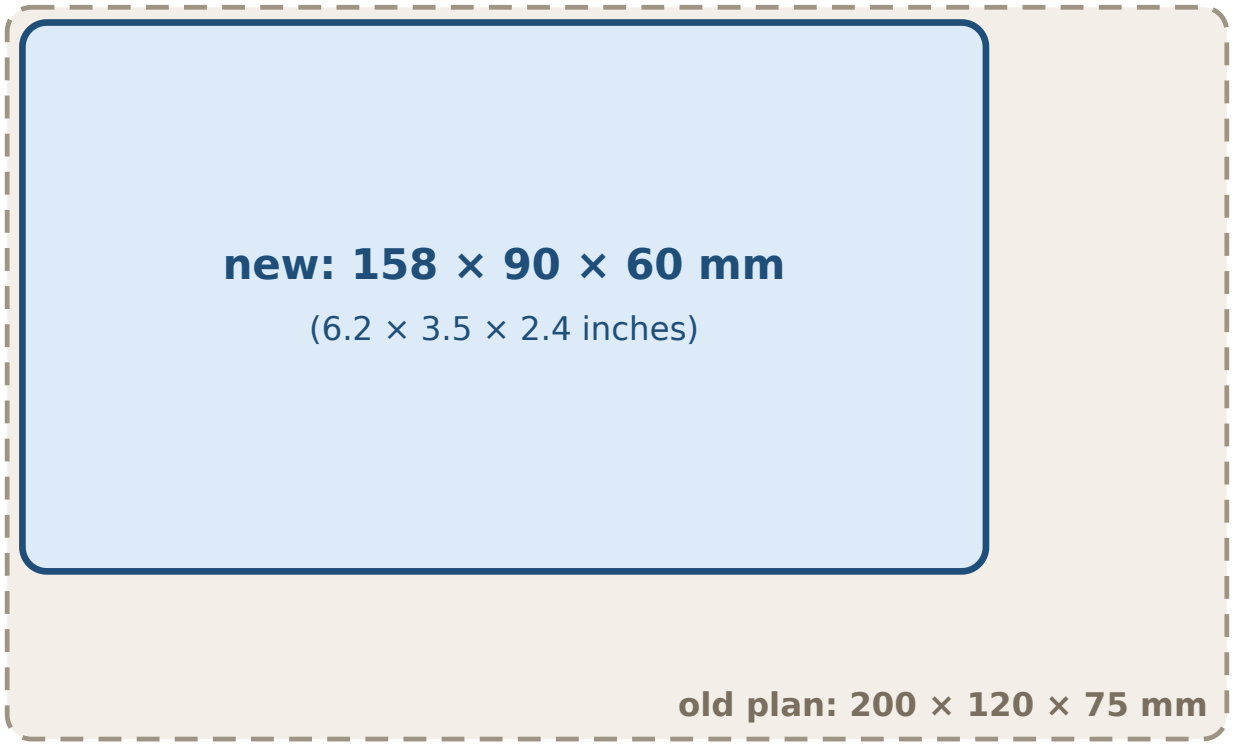
Compact Awning Box

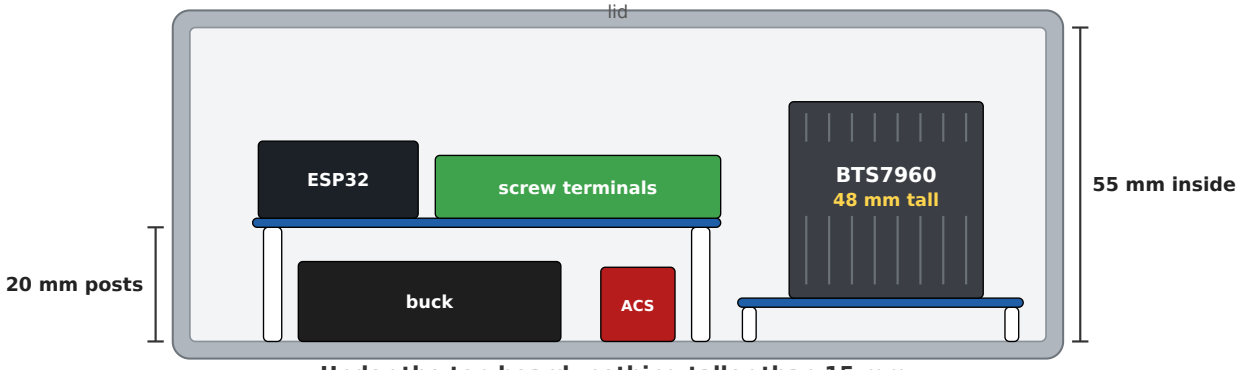
The same awning controller, stacked into a box about half the size, to fit on the awning arm. November 2026.

Same parts, about half the box



The idea in one picture

Side view: what sits on top of what



Under the top board: nothing taller than 15 mm.
The motor driver stands alone with 7 mm of air above it.

- **Level 1, the floor:** the low parts: buck converter, current sensor, perfboard, tilt sensor and two lever nuts.
- **Level 2, on four 20 mm posts:** your HCDC terminal board with the ESP32 on it, the same board as on the test rig.
- **Beside the stack:** the motor driver. It's 42 mm tall, so nothing goes over it.

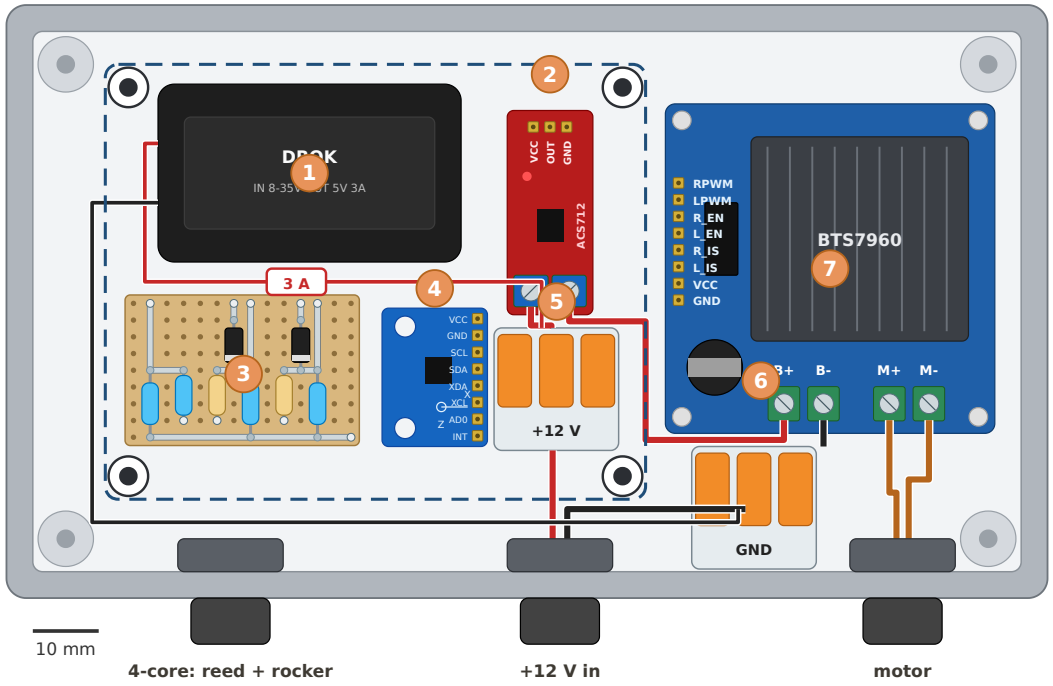
Box to buy

What	Size	Notes
IP65 junction box, ABS, clear or grey lid	**158 × 90 × 60 mm** outside (6.2 × 3.5 × 2.4 in)	Search "IP65 junction box 158x90x60". Inside should be about 150 × 82 × 55 mm.
M3 nylon standoffs	20 mm × 4, 6 mm × 8	20 mm for the top board; 6 mm for the motor driver and the perfboard
WAGO 221-413 lever nuts	2	One for +12 V, one for ground
Twin-wire ferrules	for 2 × 22 AWG	Two wires in one screw terminal (wires 11 and 12)
Inline fuse holder + 3 A fuse	mini blade	In the buck converter's +12 V lead
VHB double-sided tape	small roll	Holds the buck converter to the floor

> **Measure before you buy.** I drew your HCDC board at 82 × 66 mm, which is a little bigger than it looks in your photo, to be safe. If your board, your buck converter or the motor driver is bigger than shown here, tell me and I'll redo the drawing.

Level 1: the floor

Level 1: the box floor (looking down, lid and top board off)



Blue dashed line = where the top board sits, on 4 posts (black circles)

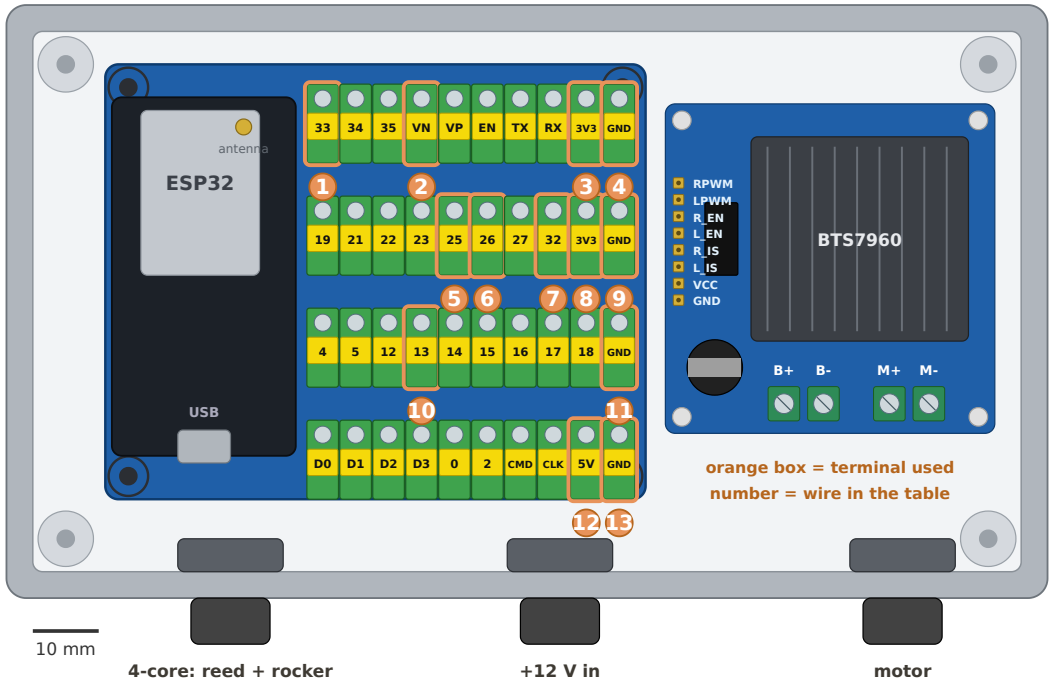
	Part, and how it's held
1	**DROK buck 12 V → 5 V** stuck down with VHB tape
2	**ACS712 current sensor** stood on end, screws down
3	**Perfboard (Figure 4b of the guide)** on 6 mm posts
4	**GY-521 tilt sensor** screwed flat with 2 screws, must not wobble
5	**Lever nut: +12 V** WAGO 221-413, zip-tied
6	**Lever nut: GND** WAGO 221-413, zip-tied
7	**BTS7960 motor driver** on 6 mm posts, heat sink up

> **The tilt sensor goes on the floor**, the stiffest place in the box, so its readings stay steady. It can face any direction: the "Save Closed / Open Position" buttons calibrate it afterwards.

Drawn to scale. Thick wires are the 12 V power wiring, which all stays on the floor. The thin wires go up to the top board.

Level 2: the top board

Level 2: the top board, 20 mm above the floor



#	From HCDC	To
1	**33**	tilt sensor SDA
2	**VN**	perfboard GPIO39
3	**3V3 (row 1)**	tilt sensor VCC
4	**GND (row 1)**	tilt sensor GND
5	**25**	perfboard GPIO25
6	**26**	perfboard GPIO26
7	**32**	reed switch (4-core)
8	**3V3 (row 2)**	driver VCC + R_EN + L_EN
9	**GND (row 2)**	driver GND
10	**13**	tilt sensor SCL
11	**GND (row 3)**	perfboard GND + ACS712 GND
12	**5V**	buck OUT+ + ACS712 VCC
13	**GND (row 4)**	buck OUT −

Rows are counted from the top of the board. Wires 11 and 12 are two wires in one terminal: use a twin ferrule. Wire 8: bridge the driver's VCC, R_EN and L_EN pins together.

Other thin wires (on the floor)

From	To	Wire
perfboard RPWM / LPWM	BTS7960 RPWM / LPWM	22 AWG
ACS712 OUT	perfboard ACS pad	22 AWG
4-core: reed GND	perfboard GND pad	4-core
4-core: rocker EXTEND / RETRACT	perfboard EXT / RET pads	4-core

Wires 1-13 are 22 AWG, except wire 7 (the reed switch), which comes in on the 4-core cable.

Power wiring on the floor

From	To	Wire
+12 V in (gland)	+12 V lever nut	14 AWG
12 V − in (gland)	GND lever nut	14 AWG
+12 V lever nut	ACS712 IP+	14 AWG
ACS712 IP−	BTS7960 B+	14 AWG
BTS7960 B−	GND lever nut	14 AWG
BTS7960 M+ / M−	motor (gland)	14 AWG
+12 V lever nut → **3 A inline fuse**	buck IN +	20 AWG
buck IN −	GND lever nut	20 AWG

> **One change from the guide:** the tilt sensor (GY-521) now runs from **3V3** instead of 5V, because the HCDC board has only one 5V terminal. The GY-521 works fine on 3.3 V. Everything else is wired exactly as in the guide.

Build order

1. **Drill the three glands** in one long side: 4-core cable, +12 V in, motor. When the box is on the arm, that side should face the knee and point down when the awning is open.
2. **Floor parts first:** motor driver and perfboard on 6 mm posts, the tilt sensor screwed flat, and the buck converter on VHB tape.
3. **Do all the floor wiring now,** while you can reach it: the thick 12 V wires, the 3 A fuse, and the lever nuts.
4. **Add the thin wires that go up** (table above). Cut them about 10 cm long and put a small tape label on each end.
5. **Screw in the four 20 mm posts** and set the HCDC board on them.
6. **Connect the labeled wires** to the HCDC terminals. Use ferrules, and use the twin ferrules for wires 11 and 12.
7. **Meter check** before the motor runs: same as page 3 of the test rig guide.
8. **Tape the ESP32's antenna wire to the inside of the lid,** away from the motor driver and the metal awning arm. That gives the best Wi-Fi.

> **Before you drill the arm,** hold the box in place and fold the awning closed. The box must clear the rail, the inner arm and the gas strut.